

H2BENCH · CALCULATION NOTE

H2Bench sizing calculations

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried into the model and sizing.py: 10 mm deck, supply beside the bench, deoxidizer (line 19), aluminium tank; R13 met, R9 at risk, cost USD 1056; oxygen crossover added as a tank safety item in section 8
0.5	2026-10-02	Amish Chadha	Decisions of 2026-10-02 noted under R8, R9 and R13; figures unchanged until the model and sizing.py are updated
0.4	2026-10-01	Amish Chadha	Constructable design (HBN-DDR-003); mass from the model's parts, cost against the value-engineering target, guard clearance corrected
0.3	2026-09-26	Amish Chadha	Budget approved by Amish (\$885); R14 from not met to met
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (electrolyzer and fill, inventory and pressure, fuel

cell, energy balance, measurement
uncertainty, hood and venting,
interlock, water and drier, envelope
and mass, cost, logging)

H2Bench sizing calculations

On paper, H2Bench meets eleven of its sixteen requirements, and cost (R14) is reported against its value-engineering target. **Three are at risk:** the H2Guard interlock (R8), hydrogen purity at the fuel cell (R9, now with a catalytic deoxidizer fitted but no supplier data) and electrolyzer back-pressure (R10, 332 kPa). Mass (R13) is met at 24.4 kg for the bench as moved. Value-engineering target USD 885; estimated cost of the constructable design USD 1056 (USD 171 over the target). Version 0.6 carries the decisions of 2026-10-02 into the model: the deck is 10 mm, the bench supply stands on the lab table beside the bench, a catalytic deoxidizer (BOM line 19) sits between the separator and the drier, and the tank is a certified aluminium cylinder. Mass falls from 28.0 to 27.4 kg with the supply (24.4 kg for the bench as moved), cost rises from USD 999 to USD 1056, R9 moves from not met to at risk and R13 from at risk to met. Version 0.4 follows the constructable design of HBN-DDR-003: the four posts, top frame, mounts and water lines added for construction raise the mass from 25.3 to 28.0 kg and the cost from USD 885 to USD 999; the made parts are now weighed from their model volumes; and the guard clearance is corrected from 18 to 50 mm. No energy, gas, pressure or venting figure changes. In v0.3, R14 was counted as met against the USD 885 budget; it is now reported as over or under the value-engineering target. Version 0.2 applies the decisions Amish accepted on 2026-09-25 (HBN-DDR-002): the relief valve moves from 350 to 325 kPa gauge, which brings R5 from at risk to met (full-lift inventory 10.5 L to 9.9 L); a pressure switch cuts the electrolyzer supply at 310 kPa gauge, so the relief valve is a backup only; and `budget_usd` rises from \$450 to \$850 with the bench supply in the kit. The switch and its relay add \$20 and 0.15 kg. Setup time (R16) needs hardware to verify. The TRL 2 energy numbers hold: a 17.7 min fill and a 29.0 min discharge return 5.2 Wh of 20.8 Wh, a round trip of 25.1 % on the HHV basis, and students can measure each stage to within 1.8 %.

Every number here is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the tank and bench dimensions and the made parts' volumes from `cad/src/model.py` and the costs from `bom/bom.csv`. All values are first-principles estimates; nothing is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Gas constants	R = 8.314 J/(mol K), F = 96,485 C/mol, 24.05 L/mol at 20 °C and 101.3 kPa; H ₂ HHV 285.83 kJ/mol, LHV 241.83 kJ/mol, γ = 1.41	Standard values; ideal gas, with a 0.24 % compressibility correction noted at 400 kPa absolute
Electrolyzer	4 cells, 1.95 V per cell, 9 A, Faraday efficiency 95 %; 2.20 V per cell at end of life	HBN-PRC-001; generic 56 cm ² class cells
Tank	2.000 L internal (inner radius 52 mm, 200.8 mm straight, hemispherical top), 3 mm wall; certified aluminium cylinder (6061-T6 class), decided 2026-10-02	<code>cad/src/model.py</code>
Gas volume outside the tank at tank pressure	0.15 L (separator headspace, drier, lines)	Estimate
Pressures	Fill 70 to 300 kPa gauge; supply cut by a pressure switch at 310 kPa gauge; relief 325 kPa gauge with 10 % overpressure at full lift; check valve 7 kPa; kept at 20 kPa gauge overnight	HBN-PRC-001 v0.4, HBN-DDR-002, typical relief valve
Temperatures	Lab 20 °C; coldest 15 °C; separator gas 25 °C	HBN-PRB-001
Storage loss	2 % of the hydrogen made per cycle	Estimate, to be measured
Fuel cell	13 cells, 0.60 V at 1.5 A (7.8 V, 11.7 W), fuel utilization 95 %, 0.9 W fan and controls; 0.18 L/min at full output; 45 to 55 kPa gauge supply; 99.995 % purity	Listing for the 12 W class (Fuel Cell Store , checked 2026-09-25)
Measurement	Transducer 0.25 % of 600 kPa full scale; thermistor ± 1 K; tank volume ± 1 % by water fill; power ± 1 %	HBN-REQ-001 R1
Room and hood	30 m ³ room; 60 m ³ /h through the canopy duct	Assumed hood share of H2Guard's 150 m ³ /h continuous exhaust (HGD-PRC-001)
H2Guard	Warning 0.4 % vol (10 % LFL), trip 1.0 % vol (25 % LFL); trip to valve closed 1.1 s	HGD-REQ-001 R2 to R4, HGD-PRC-001 (TRL 2 estimates)
Drier	50 g silica gel, 10 % usable water uptake	Typical indicating gel at low humidity
Value-engineering target	USD 885 (<code>budget_usd</code>), bench supply included, H2Guard excluded; a hypothetical control target, not a limit	HBN-DDR-002; Amish, 2026-10-01

■ 2. Electrolyzer and fill (R2, R3, R10)

The stack draws $7.8 \text{ V} \times 9 \text{ A} = \mathbf{70.2 \text{ W}}$ (79 W with aged cells at 2.20 V), inside the 100 W limit (R3) and the 30 V, 10 A supply. Faraday's law gives $1.772 \times 10^{-4} \text{ mol/s}$, or **256 mL/min** at 20 °C. Filling the 2.000 L tank from 70 to 300 kPa gauge takes 0.1887 mol (4.54 L at 20 °C and 101.3 kPa), so the fill lasts **1,065 s (17.7 min)** and uses **20.77 Wh**. The hydrogen holds 14.99 Wh (HHV): an electrolyzer efficiency of **72.2 %** (voltage efficiency 75.9 %, Faraday efficiency 95 %). The stack gives off about 19.5 W of heat.

The electrolyzer compresses its own hydrogen. The Nernst term at 300 kPa gauge is **17.4 mV per cell**, 0.89 % of the cell voltage, so no compressor is needed.

Back-pressure (R10). The membrane must hold the tank pressure plus the check valve cracking pressure over the oxygen side: **307 kPa** in normal filling and **317 kPa** at the 310 kPa gauge supply cut. The supply cut stops the stack before the relief valve can lift, so the relief is a backup only; if the cut failed, the membrane would see **332 kPa** at the 325 kPa gauge relief setting and 365 kPa at full lift. R10 is restated as 332 kPa or more (HBN-REQ-001 v0.4, was 300 kPa). No generic stack rating is in hand. At risk.

■ 3. Inventory and pressure (R5, R6, R7)

Table 2. Hydrogen held, expressed at 20 °C and 101.3 kPa.

State	Hydrogen
Tank at 300 kPa gauge	0.3293 mol, 7.92 L, 0.664 g
Tank and lines at 300 kPa gauge	8.52 L
Tank and lines at relief set, 325 kPa gauge	9.05 L (9.20 L at 15 °C)
Tank and lines at full relief lift (358 kPa gauge) and 15 °C	9.90 L

The TRL 2 note gave 7.9 L "at the relief setting"; that is the tank alone at the working pressure. With the relief at 350 kPa gauge (v0.1) the full-lift case on a cold day was 10.5 L. A relief set at 329 kPa gauge or less keeps even that case under 10 L, and 325 kPa gauge is now decided (HBN-DDR-002): 9.0 L at the relief set point and **9.9 L** at full lift and 15 °C. **R5 is met**, with a 0.1 L margin in the worst case.

Pressure parts (R6). Working 300 kPa gauge, supply cut 310 kPa gauge, relief 325 kPa gauge. The modeled vessel (110 mm outside diameter, 3 mm aluminium wall) has a hoop stress of 17.8 MPa at its 1 MPa rating and 71.3 MPa at four times that, against a 6061-T6 yield of 276 MPa. Stored pressure energy is **646 J** at 300 kPa gauge (710 J at the 325 kPa gauge relief setting), by isentropic expansion to atmosphere. Stored chemical energy is 94 kJ (HHV), 80 kJ (LHV). Met.

Room release (R7). The relief-set inventory mixed into a 30 m³ room gives **0.030 % by volume, 0.75 % of the lower flammable limit**; the worst case is 0.033 %. Met by a wide margin. H2Guard's proposed inventory rule (1 % of room volume, 300 L for 30 m³, HGD-REQ-001 R9) is met thirtyfold.

■ 4. Fuel cell and discharge (R2, R4)

Thirteen cells at 0.60 V and 1.5 A give **11.7 W gross and 10.8 W net** after 0.9 W for the fan and controls, which meets R4 with a 0.8 W margin. At 0.58 V per cell the net is 10.4 W; at 0.55 V it falls to 9.8 W, below target. The stack reacts 146 mL/min and, at 95 % utilization, draws **154 mL/min**, inside the listed 0.18 L/min at full output. (The TRL 2 table's 146 mL/min was the reacted flow.)

After 2 % storage loss, 0.1850 mol is available, which runs the fuel cell for **1,739 s (29.0 min)** and delivers **5.22 Wh**. Net fuel cell efficiency is 35.5 % (HHV) or 42.0 % (LHV), consistent with the 40 % system efficiency listed for the 12 W class. About 18.7 W of heat and purge loss leaves the stack. The full cycle runs for **46.7 min**, inside the 60 min of R2. Met.

■ 5. Energy balance

Table 3. Energy per lesson cycle, HHV basis. These values drive [media/flow.png](#).

Stage	Energy	Loss
Electricity in	20.8 Wh	
Hydrogen made	15.0 Wh	Electrolyzer heat 5.8 Wh
After storage	14.7 Wh	Leaks and purges 0.3 Wh
Fuel cell net DC	5.2 Wh	Fuel cell heat, purge and fan 9.5 Wh

The round trip is **25.1 %** (HHV basis), just below the 27.4 to 48 % range published for power-to-hydrogen-to-power ([Oxford Institute for Energy Studies, 2025](#)), as expected at 12 W scale. Each fill splits 3.40 g of water and vents 2.27 L of oxygen.

■ 6. Measurement uncertainty (R1)

Hydrogen is counted from pressure and temperature before and after each stage: $n = PV/(RT)$.

Table 4. Standard uncertainty ($k = 1$) of the hydrogen made per fill.

Source	Contribution
Pressure, 1.5 kPa per reading, two readings	0.92 %
Temperature, ± 1 K, two readings	0.65 %
Tank volume, ± 1 %	1.0 %
Combined	1.51 %

The thermistor is bonded to the tank wall, so the gas temperature matters. During the fill the flow work into the tank heats the gas about 0.95 K above the wall, and the gas settles to the wall with a 15 s time constant; readings are taken 2 min after the fill ends. With power measured to ± 1 %, the stage efficiencies carry: electrolyzer 1.8 %, storage 1.0 %, fuel cell 1.8 %, round trip 1.4 % and Faraday efficiency 1.8 %. All are within the ± 5 % of R1. Met.

■ 7. Hood, venting and interlock (R8, R15, R16)

Hood. At 60 m³/h (1,000 L/min) the 100 mm duct runs at 2.12 m/s. The electrolyzer's full output leaking into the hood gives only 0.026 % by volume at the duct. A leak must exceed 4.0 L/min to reach H2Guard's warning at the duct and 10 L/min to trip it. Small leaks are diluted by the extraction; the sensor at the high point catches large ones and a stopped fan (HGD-REQ-001 R5).

Venting. Venting the tank and lines from 300 to 20 kPa gauge releases 5.94 L. A needle valve set for 3.0 min (an effective orifice of about 0.16 mm) gives a peak of 3.50 L/min, or 0.35 % by volume at the duct (8.7 % of LFL), just under H2Guard's 0.4 % warning. At a constant flow, a vent shorter than 89 s would reach the warning. The end-of-day vent after a discharge (70 to 20 kPa gauge) is only 1.06 L. The 3.0 min vent meets the 5 min shutdown part of R16; setup time needs a walk-through with hardware, so **R16 is not verifiable at TRL 3**.

Interlock (R8). From H2Guard's TRL 2 figures, the trip reaches a closed valve and a dead supply in about 1.1 s (comparator and relay 0.1 s, solenoid 1.0 s), inside 2 s; the electrolyzer makes only 4.7 mL of hydrogen in that time. But H2Guard is itself at TRL 2, its valve is a 24 V part while H2Bench's tank solenoid is 12 V, and whether its relay

can break the 9 A DC supply line is not specified. **At risk.** On 2026-10-02 Amish decided to propose to H2Guard a 24 V normally closed tank solenoid powered directly from H2Guard, its alarm contact in series with the 310 kPa cut relay, and 60 m³/h stated as the hood extraction rate, separate from H2Guard's 150 m³/h room exhaust.

Guard (R15). The four guard rods stand at the corners of a 156 mm square, so each clears the tank shell by 50 mm (v0.3 gave 18 mm, measured as if the rods stood on the axes); the top plate stops the tank lifting and the cradle pocket stops it tipping. The relief and vent lines run to the outlet hanger under the canopy in the model. Met by design review.

■ 8. Water, drier and purity (R9, R11)

Gas leaving the separator at 25 °C carries 0.79 to 1.85 % water vapour by mole, depending on tank pressure: about 0.045 g of water per fill. 50 g of silica gel at 10 % uptake lasts about **110 fills** before its color changes. The drier removes water, but PEM stacks also pass some oxygen across the membrane, and the fuel cell listing asks for 99.995 % or better, which is 50 ppm of all impurities together. On 2026-10-02 Amish decided to fit a catalytic deoxidizer between the separator and the drier (BOM line 19, USD 60, 0.16 kg), and to remove it only if the stack supplier's data show oxygen within the 99.995 % limit at the bench's lowest operating current. The cartridge reacts oxygen with hydrogen to water over a palladium catalyst. Because the crossover is unknown, three cases are run: at 0.1, 0.5 and 2.0 % oxygen in the hydrogen, the cartridge reacts 0.26, 1.28 and 5.12 mL/min of oxygen, warms by 0.09, 0.43 and 1.71 W, and forms 7, 34 and 136 mg of water per fill (the drier takes 5 g). At an assumed 99 % conversion the residual oxygen is 10, 50 and 200 ppm, so the limit is met only at the low end; **R9 is at risk**, not met, until the supplier gives the oxygen figure and the cartridge's conversion. Action: ask the stack supplier for oxygen in the hydrogen at the bench's lowest operating current (outreach by Amish, open).

Oxygen crossover as a tank safety item. Oxygen in the stored hydrogen is a safety question for the tank as well as a purity question for the fuel cell. A hydrogen and oxygen mixture is flammable over a wide range, with a lower limit of about 4 % oxygen in hydrogen (handbook value, to confirm), and the tank is a closed vessel at up to 325 kPa gauge with a pressure energy of 710 J at relief. At 2.0 % crossover before the deoxidizer the mixture is within a factor of two of that limit; after a 99 % conversion it is 0.02 %, far below it. The deoxidizer is therefore the barrier that keeps the tank gas outside the flammable range, and it must not be removed on a purity argument alone: removal should wait for the supplier's oxygen figure at the lowest operating current, and the margin below the 4 % limit at the tank is for Amish to set (proposed: at least tenfold). The check valve and flame arrestor sit after the deoxidizer and the drier, so a flame cannot travel back into the stack. Until the figure is in hand, the operating rules stay: pressure 300 kPa gauge or less, guard in place, H2Guard running.

Water quality (R11) is met by the mixed-bed resin and conductivity check.

■ 9. Envelope and mass (R13)

The model's envelope above the lab table is **900 x 450 x 753 mm**, inside 1,000 x 500 x 800 mm.

Table 5. Mass estimate. Made parts are weighed from their volumes in `cad/src/model.py` (aluminium 2,700 kg/m³, HDPE 950 kg/m³, polycarbonate 1,200 kg/m³, extrusion 0.5 kg/m); bought parts use typical catalogue masses.

Item	Mass (kg)
1 Frame (3.44 m of extrusion, 8 brackets), deck (10 mm, notched), drip lip	1.88 + 3.81 + 0.12
2 Back panel, 3 mm aluminium composite, 900 x 550 mm	1.88
3 Posts and top frame (4.65 m of extrusion, 14 brackets)	2.60
3 Canopy, duct collar and outlet hanger	1.30
4 Bench power supply (stands beside the bench, not on it)	3.00
5 Reservoir (filled), deionizer, stand, water post and clips	0.80 + 0.29
6 Electrolyzer stack and feet	3.00 + 0.06
7 Separator, drier, column bracket and clips	0.40 + 0.17
19 Catalytic deoxidizer cartridge and clip (assumed)	0.16
8 Check valve, arrestor, saddle and clip	0.32
9 Tank shell (aluminium, from the model); cradle, guard rods, nuts and top plate	0.80 + 1.31
10 and 11 Manifold with cut switch, regulator and solenoid	1.30
12 Fuel cell (275 g listed) with controller, and its bridge	0.40 + 0.06
13 and 14 Load, lamp, meters, display and cut relay	0.75
15 H2Guard parts on the bench	1.50
16 and 17 Tubing (gas and water lines), wiring, fasteners and T-nuts	1.50
Total	27.4 (24.4 for the bench as moved, without the supply)

The envelope footprint is unchanged and the height falls from 755 to 753 mm with the 10 mm deck. The mass was 28.0 kg with the supply at v0.5 (the rear posts, top frame and brackets added about 2.4 kg to the 25.3 kg of v0.3). The decisions of 2026-10-02 change it: the 10 mm deck saves 0.77 kg, the deoxidizer adds 0.16 kg, and the bench supply stands beside the bench, so the bench as moved is **24.4 kg**, inside the 25 kg limit with 0.6 kg to spare; with the supply it is 27.4 kg. **R13 is met**, provided the tank stays aluminium: a stainless tank of the same wall would add about 1.6 kg and break the limit (stainless is allowed only at about 1.0 kg or less, HBN-DEC-001).

■ 10. Cost (R14)

Value-engineering target: USD 885 (budget_usd , a hypothetical control target, not a limit; Amish, 2026-10-01). Estimated cost of the constructable design: USD 1056 (USD 171 over the target), H2Guard excluded. Without the bench supply it is USD 991 (USD 106 over), and without the supply and the plug-in RCD USD 971 (USD 86 over). Since v0.5 the catalytic deoxidizer (BOM line 19) adds USD 60 (an estimate with no quote: a small palladium cartridge, two compression fittings and a clip) and the 10 mm deck saves USD 3 (HDPE sheet bought by weight at about USD 4 per kg, 3.80 kg against 4.57 kg), so USD 999 becomes USD 1056. The parts added for construction

(HBN-DDR-003) account for USD 114 of the rise from USD 885, across BOM lines 1, 3, 5, 6, 7, 8, 9, 12, 16 and 17. The two stacks with their mounts are USD 408, 39 % of the total. The main cost drivers and the savings worth trying are in the design decisions register (HBN-DEC-001, Value engineering).

■ 11. Logging (R12)

One cycle is 2,804 s, so a 1 Hz log with 10 channels is 2,804 rows, about 224 kB of CSV. Met by design review.

■ 12. Results

Table 6. Results against HBN-REQ-001 v0.8. Written to `docs/04-calcs/results.csv`.

ID	Value	Target	Status
R1	Worst stage 1.8 % (hydrogen per fill 1.5 %, power 1 %)	Each stage within ± 5 % relative	Met
R2	17.7 min fill + 29.0 min discharge = 46.7 min	60 min or less	Met
R3	70.2 W (79 W with aged cells)	100 W or less	Met
R4	10.8 W net (11.7 W gross); 9.8 W if cells fall to 0.55 V	10 to 50 W	Met
R5	8.5 L at 300 kPa gauge; 9.0 L at relief set; 9.9 L at full relief lift and 15 °C	10 L or less	Met
R6	300 kPa gauge working, relief 325 kPa gauge; hoop stress 18 MPa at 1 MPa	300 and 350 kPa gauge or less; vessel 1 MPa or more	Met
R7	0.030 % by volume (0.8 % LFL); 0.033 % worst case	Below 25 % LFL in 30 m ³	Met
R8	1.1 s trip to valve closed, from H2Guard TRL 2 figures; 24 V and 12 V valve mismatch	Supply cut and solenoid closed within 2 s; fan runs	At risk
R9	Water removed by the drier; deoxidizer fitted; residual oxygen 10 to 200 ppm at an assumed 99 % conversion against 50 ppm for all impurities; no supplier data	99.995 % or better	At risk
R10	Needs 307 kPa in filling, 317 kPa at the supply cut, 332 kPa at relief set; no rating in hand	332 kPa or more over the O ₂ side	At risk
R11	Mixed-bed resin with conductivity check; 3.4 g water per fill	1 μ S/cm or less	Met
R12	2,804 rows per cycle, about 224 kB	V, I, P, p and T at 1 Hz to CSV	Met
R13	900 x 450 x 753 mm; 24.4 kg for the bench as moved, supply beside it (27.4 kg with the supply)	1,000 x 500 mm, 800 mm, 25 kg or less (bench as moved)	Met
R14	USD 1056 full; USD 991 without the supply; USD 971 minimum (H2Guard excluded)	USD 885 value-engineering target, supply included	USD 171 over the value-engineering target
R15	Guard clearance 50 mm; relief and vent piped to the outlet hanger under the canopy	Guard, piped relief, no tool-free fitting	Met
R16	Vent 3.0 min; setup needs a walk-through with hardware	Setup 15 min or less; safe in 5 min or less	Not verifiable at TRL 3

■ 13. Safety

Safety: Hydrogen is flammable from about 4 to 74 % in air and ignites with about 0.02 mJ. These calculations bound the inventory (9.9 L worst case) and the concentration at the duct, but they do not replace H2Guard's detection and interlock, supervision, or a site risk assessment. H2Bench is a research and teaching prototype, not certified laboratory equipment.

Safety: The tank and lines hold up to 710 J of pressure energy at the relief setting. The 310 kPa gauge supply cut keeps the relief valve as a backup; test the cut before each lesson series. Use only vessels and fittings rated 1 MPa or more for hydrogen, keep the guard in place and never open a pressurized fitting.

Safety: Oxygen carried across the electrolyzer membrane can build up in the stored hydrogen. Keep the catalytic deoxidizer in line until the stack supplier's oxygen figure at the lowest operating current shows a wide margin below the flammable limit (section 8).

Safety: Vent slowly. Venting a full tank in under about 90 s can raise the concentration at the duct past H2Guard's warning level; the needle valve is set for 3 min or more.